

1. Key Differences Between Wireless Communications and Cellular Communications

✅ Wireless Communications

According to the slide:

“Wireless communications is a type of data communication that is performed and delivered wirelessly.”

◆ Definition:

Wireless communication is a **broad concept** that includes all communication systems where signals are transmitted without physical wires using electromagnetic waves.

◆ Key Characteristics:

- Communication without cables
- Can be short-range or long-range
- Includes many technologies:
 - WiFi (IEEE 802.11)
 - Bluetooth
 - ZigBee
 - Satellite systems
 - Cellular systems
 - mmWave systems
 - Ad-hoc networks

◆ Modern Usage:

- WiFi for local internet access
- Bluetooth for cable replacement
- ZigBee for IoT sensors
- Satellite for GPS and broadcasting
- mmWave for high-speed Gbps transmission
- IoT networks for smart homes, health monitoring, etc.

👉 **Wireless communication is the umbrella technology.**

✅ **Cellular Communications**

From the slide:

“A cellular network or mobile network is a communication network where the last link is wireless. The network is distributed over land areas called cells.”

◆ **Definition:**

Cellular communication is a **specific type of wireless communication** that divides geographic areas into **cells**, each served by a **base station**.

◆ **Key Characteristics:**

- Infrastructure-based network
- Uses base stations (cell towers)
- Supports handover between cells
- Designed for mobility
- Frequency reuse for capacity improvement

◆ **Evolution (from slides):**

- AMPS (analog)
- GSM (2G)
- UMTS (3G)
- LTE (4G)
- 5G (Emerging)

◆ **Modern Usage:**

- Mobile phones
- 4G/LTE internet
- 5G networks
- Voice + data + multimedia

- Indoor/outdoor wide-area coverage
-

Major Differences

Feature	Wireless Communication	Cellular Communication
Scope	Broad category	Subset of wireless
Infrastructure	May or may not need infrastructure	Requires base stations
Coverage	Local to global	Large-area structured coverage
Mobility Support	Limited in some systems	Full mobility + handover
Example	WiFi, Bluetooth	LTE, 5G

Summary:

- **All cellular systems are wireless**
 - **Not all wireless systems are cellular**
 - Cellular networks are structured, wide-area, mobility-focused systems.
 - Wireless includes everything from IoT sensors to satellites.
-

2(i) Are We Operating at the Shannon Limit? What Would Shannon Say?

From the PPT:

“We’re 99% of the way to the barrier known as Shannon’s limit.”

“We don’t know the Shannon capacity of most wireless channels.”

“Time-varying channels.”

“Channels with interference or relays.”

Are we at the Shannon Limit?

Short Answer: No, not completely.

Although modern systems (like LTE and WiFi 802.11ac) are highly efficient, we are:

- Close for simple channels

- Not at the limit for complex wireless environments

Why?

Because:

- Wireless channels are **time-varying**
- They contain **interference**
- They involve **multi-user systems**
- There are **delay, energy, and cost constraints**
- Cellular systems have complex topology

The slide clearly says:

“We don’t know the Shannon capacity of most wireless channels.”

So we cannot say we are fully at the limit.

What Would Shannon Say?

Claude Shannon would likely say:

1. Capacity depends on channel model.
2. Many real wireless channels do not have known capacity formulas.
3. Interference and relays complicate capacity.
4. There is still room for innovation.

The slide also says:

“Shannon theory provides design insights and system performance upper bounds.”

Meaning:

- Shannon gives the theoretical maximum.
- Real systems are constrained by hardware, energy, interference.

So Shannon would say:

👉 “You are close in simple cases, but real-world wireless systems are much more complex.”

2(ii) Why Does WiFi Performance Reduce? How Can We Overcome It?

Why WiFi performance reduces (From slide):

- “Carrier Sense Multiple Access: if another WiFi signal detected, random backoff.”
 - “Collision Detection: if collision detected, resend.”
 - “The WiFi standard lacks good mechanisms to mitigate interference, especially in dense AP deployments.”
 - “Static channel assignment, power levels, and carrier sensing thresholds.”
 - “WiFi systems exhibit poor spectrum reuse and significant contention among APs and clients.”
 - “Result is low throughput and a poor user experience.”
-

How to Overcome (From slide):

- “Multiuser MIMO will help each AP.”
 - “SoN-for-WiFi: dynamic self-organization network software to manage WiFi APs.”
 - Channel Selection
 - Power Control
 - “Software-Defined Wireless Networks.”
 - “Cloud-based software to dynamically control and optimize wireless networks.”
 - mmWave systems (60–80 GHz, massive MIMO, Gbps data rates).
-

Final Summary

1 Wireless vs Cellular

- Wireless = broad concept
- Cellular = structured mobility-based wireless system

2 Shannon Limit

- We are close in simple channels

- Not fully at the limit in real-world wireless
- Capacity unknown for many wireless systems

3 WiFi Performance Reduction

Due to:

- CSMA backoff
- Collisions
- Interference
- Static configuration
- Dense deployments

Solutions:

- Multiuser MIMO
- SoN
- SDWN
- mmWave
- Better spectrum control

1. Compare theoretical (and often announced) values for transmission data rates with realistic data rates. What could be reasons for the sometimes big difference? How do fixed and wireless systems differ in this context?

1. Theoretical vs. Realistic Transmission Data Rates

Theoretical (Announced) Data Rates

These are:

- Peak PHY layer rates under ideal conditions
- Based on maximum bandwidth, highest modulation/coding schemes, full MIMO streams
- Assume no interference, perfect channel conditions, no protocol overhead

Examples from the PPT:

- 3G peak: 384 Kbps
- 4G/LTE: 50–100 Mbps
- 802.11ac: up to several Gbps
- WiGig: 7–25 Gbps

These represent **maximum physical-layer bit rates**, not user throughput.

Realistic Data Rates

Actual user data rates are much lower due to:

Physical Layer Limitations

- Path loss
- Shadowing
- Multipath fading
- Interference from other users
- Mobility and time-varying channels
- Distance from base station/AP

Protocol & Network Overheads

- MAC layer overhead (CSMA/CA in WiFi)
- Control signaling
- Retransmissions due to errors
- Guard intervals
- Packet headers
- Scheduling delays
- Backhaul limitations

Shared Medium Effects

- Contention among users
- Random backoff (WiFi)

- Cell congestion (cellular systems)

Shannon Limit Considerations

The PPT notes that we approach theoretical limits (Shannon capacity), but:

- Real channels are time-varying
 - Interference-limited environments dominate
 - Practical constraints (energy, delay, cost) apply
-

Why the Difference Can Be Large

The gap is large because:

- Theoretical rates assume **ideal SNR**
 - Real systems operate in **interference-limited** conditions
 - Resources are **shared**
 - Environmental conditions are unpredictable
 - Hardware imperfections reduce performance
 - Adaptive modulation often drops to lower rates in poor conditions
-

Fixed vs Wireless Systems

Aspect	Fixed (Wired)	Wireless
Channel Stability	Very stable	Time-varying
Interference	Minimal	Significant
Bandwidth Sharing	Controlled	Shared medium
Path Loss	Negligible	Distance-dependent
Fading	None	Severe multipath
Achievable Rate Gap	Small	Large

Key Difference:

Wireless systems are **interference-limited and channel-variable**, whereas fixed systems are **noise-limited and stable**. Therefore, wireless shows a much larger gap between theoretical and practical rates.

2(iv). Fundamental Causes of Wi-Fi Performance Degradation

From the PPT discussion on WLANs and WiFi:

A. Physical Layer Causes

1. Interference (co-channel & adjacent channel)
 2. Multipath fading
 3. Path loss
 4. Limited spectrum (especially 2.4 GHz band)
 5. Dense AP deployments
-

B. MAC Layer Causes

1. **CSMA/CA random access**
 - Devices must sense channel before transmission
 - Random backoff increases delay
2. Collisions and retransmissions
3. Poor spectrum reuse
4. Static channel assignment
5. Fixed power levels
6. Carrier sensing thresholds

In dense deployments:

- APs contend with each other
- High contention
- Low throughput
- Poor user experience

Multuser MIMO helps within an AP, but not between interfering APs.

2(v). Trade-offs Between PoE, AC, and Solar for Network Radios

From energy-harvesting and green network slides:

1. Power over Ethernet (PoE)

Advantages:

- Single cable for data + power
- Easy deployment
- Centralized backup via UPS
- Good for indoor APs

Disadvantages:

- Limited power budget
 - Distance limitations (~100m Ethernet limit)
 - Infrastructure dependency
-

2. AC Power

Advantages:

- High power availability
- Suitable for high-performance base stations
- Stable power supply

Disadvantages:

- Requires electrical installation
 - Higher installation cost
 - Not flexible for remote areas
-

3. Solar / Energy Harvesting

Advantages:

- Useful in remote areas
- Green/low operational cost
- Supports IoT & sensor networks
- Enables off-grid deployments

Disadvantages:

- Intermittent energy arrivals
- Storage (battery) required
- Lower reliability
- New communication design principles required

Trade-off Summary

Source	Reliability	Power Level	Deployment Ease	Cost	Sustainability
PoE	High	Moderate	Easy (indoor)	Moderate	Medium
AC	Very High	High	Less flexible	Higher	Low
Solar	Variable	Low-Moderate	Good remote	High initial	Very High

2(vi). Limitations Preventing Universal SON Implementation

SON = Self-Organizing Networks (as discussed in WiFi and SDWN slides)

Although SON promises:

- Dynamic channel selection
- Power control
- Interference mitigation
- Network analytics

It is not universally implemented due to:

1. Technical Limitations

A. Heterogeneous Networks

- Different vendors
- Different standards (WiFi, LTE, Zigbee, etc.)
- Interoperability issues

B. Dynamic Environments

- Rapid channel variation
- Mobility
- Unpredictable interference

C. Scalability

- Large networks difficult to coordinate centrally

D. Legacy MAC Protocols

- WiFi uses CSMA/CA (not designed for centralized optimization)

E. Backhaul Constraints

- SON requires coordination signaling
- Adds overhead

2. Operational & Practical Limitations

1. Cost of deployment
 2. Complexity of implementation
 3. Security concerns
 4. Reliability risks (automatic misconfiguration)
 5. Regulatory constraints
 6. Vendor competition (lack of unified standards)
-

3. Fundamental Challenge

Wireless networks are:

- Interference-limited
- Decentralized
- Highly dynamic

Universal SON would require:

- Full cross-layer visibility
- Cross-vendor standardization
- Real-time centralized/cloud intelligence (SDWN vision)

This is still an open research and engineering problem.

Alternate answer

1. Theoretical vs. Realistic Transmission Data Rates

Theoretical (Announced) Data Rates

- Peak PHY layer rates under ideal conditions
- Maximum bandwidth, highest modulation/coding schemes, full MIMO streams
- Assume no interference, perfect channel conditions, no protocol overhead

Examples:

- 3G peak: 384 Kbps
- 4G/LTE: 50–100 Mbps
- 802.11ac: several Gbps
- WiGig: 7–25 Gbps

→ Maximum physical-layer bit rates, not user throughput.

Realistic Data Rates

Much lower due to:

Physical Layer Limitations

- Path loss
- Shadowing
- Multipath fading
- Interference from other users
- Mobility, time-varying channels
- Distance from base station/AP

Protocol & Network Overheads

- MAC layer overhead (CSMA/CA in WiFi)
- Control signaling
- Retransmissions
- Guard intervals
- Packet headers
- Scheduling delays
- Backhaul limitations

Shared Medium Effects

- Contention among users
- Random backoff (WiFi)
- Cell congestion (cellular systems)

Shannon Limit Considerations

- Approach Shannon capacity
 - Real channels are time-varying
 - Interference-limited environments
 - Energy, delay, cost constraints
-

Why the Difference Is Large

- Ideal SNR assumed
 - Interference-limited conditions
 - Shared resources
 - Unpredictable environment
 - Hardware imperfections
 - Adaptive modulation drops to lower rates
-

Fixed vs Wireless Systems

Aspect	Fixed (Wired)	Wireless
Channel Stability	Very stable	Time-varying
Interference	Minimal	Significant
Bandwidth Sharing	Controlled	Shared medium
Path Loss	Negligible	Distance-dependent
Fading	None	Severe multipath
Achievable Rate Gap	Small	Large

Key Difference:

Wireless = interference-limited, channel-variable.

Fixed = noise-limited, stable.

→ Wireless has a larger gap between theoretical and practical rates.

2(v). PoE vs AC vs Solar Trade-offs**Power over Ethernet (PoE)****Advantages:**

- Single cable (data + power)
- Easy deployment
- Centralized UPS backup
- Indoor APs

Disadvantages:

- Limited power budget
 - ~100m Ethernet limit
 - Infrastructure dependency
-

AC Power**Advantages:**

- High power availability
- High-performance base stations
- Stable supply

Disadvantages:

- Electrical installation required

- Higher cost
- Not flexible for remote areas

Solar / Energy Harvesting

Advantages:

- Remote areas
- Green, low operational cost
- IoT & sensor networks
- Off-grid deployment

Disadvantages:

- Intermittent energy arrivals
- Battery storage required
- Lower reliability
- New communication design principles

Trade-off Summary

Source Reliability Power Level			Deployment Ease	Cost	Sustainability
PoE	High	Moderate	Easy (indoor)	Moderate	Medium
AC	Very High	High	Less flexible	Higher	Low
Solar	Variable	Low–Moderate	Good remote	High initial	Very High

2(vi). Limitations of Universal SON

SON (Self-Organizing Networks) promises:

- Dynamic channel selection
- Power control
- Interference mitigation
- Network analytics

Technical Limitations

Heterogeneous Networks

- Different vendors
- Different standards (WiFi, LTE, Zigbee)
- Interoperability issues

Dynamic Environments

- Rapid channel variation
- Mobility
- Unpredictable interference

Scalability

- Large networks hard to coordinate

Legacy MAC Protocols

- WiFi uses CSMA/CA
- Not designed for centralized optimization

Backhaul Constraints

- Coordination signaling required

- Added overhead
-

Operational Limitations

- Deployment cost
 - Implementation complexity
 - Security concerns
 - Automatic misconfiguration risk
 - Regulatory constraints
 - Vendor competition
-

Fundamental Challenge

Wireless networks are:

- Interference-limited
- Decentralized
- Highly dynamic

Universal SON requires:

- Cross-layer visibility
- Cross-vendor standardization
- Real-time centralized/cloud intelligence (SDWN vision)

→ Still an open research and engineering problem.